



Data Structure

IT245

Project

Deadline: Day 8/9/2026 @ 23:59

[Total Mark is 20]

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

CLO6:

Illustrate data structure problems and their designed solutions in the form of a group project.

Description and Instructions

1. Objective

The primary objective of this project is to gain proficiency in the design and development of algorithmic solutions through the application of data structures concepts.

2. Introduction

In this project, student groups will collaboratively solve a problem designed to assess their comprehension of data structures. Each group, consisting of **three to four** students, will consolidate their work into a **single report**, which will be submitted alongside the **other project materials**.

3. Required work

Select an algorithmic problem pertaining to data structures concepts (specifically one covered during class lectures) and **formulate** a comprehensive problem statement addressing it.

Examples of project topics:

Problem Name	Problem Description
Task Scheduling System	Schedule tasks while respecting priorities or arrival orders.
Route Planning System	Determine the optimal route between two locations on a given map.
Dictionary Word Lookup	Check whether words exist in a dictionary (set of sample words) and suggest corrections.

Book Inquiry System	Organize, search, and retrieve book names.
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- a. **Choose** a project from the table provided above or **propose** an original project idea.
- b. **Analyze and comprehend** the selected problem thoroughly.
- c. Implement at **least two different solutions** to the problem using data structures.
- d. At least **one data structure** must be incorporated into **each** proposed solution.
- e. Solutions must be implemented in separate methods, and they should be demonstrated and tested in a main method.
- f. Provide a **comprehensive analysis** of the **algorithmic complexity** for **both** implemented solutions.

Note: You may **choose** a project from the table above **or propose** an original idea. If you **propose** your own topic, you **must receive confirmation** from your instructor before proceeding.

4. What to submit?

Mid project Progress Report: (2 marks)

Students are required to submit a short mid progress report, which must include:

1. **Work completed this period**
2. **One key decision has been made during this period**
3. **One problem encountered**

4. Plan to do next.
5. Each group member input/participation.

Version History / Commit Log Requirements: (15 marks)

Students must complete and submit project history/timestamp using a live, editable link:

- **Written report:** Submit a shared Google Doc or Word Online link (sharing set to "Anyone with the link can view") along with your final PDF report. The link must show the progress of students' work with timestamps/screenshots.
- **Code/data projects:** Submit code via a Google Colab, GitHub, or GitLab repository link (coordinator may choose a repository suitable for the course). The repository must show commit history reflecting incremental progress. *commit regularly (at minimum once per week), with clear, descriptive commit messages.*
- **A separate section in the report must be added and named "Project Artifacts" to include all links such as:**

Project Artifacts:

- **Live report document (with edit history):** [link]
- **GitHub/GitLab repository (with commit history):** [link]
- **Google Colab notebook (if applicable):** [link]

What NOT to do: Students must not write their report elsewhere and paste the final version in at the last minute. They also should not make a single commit containing your entire codebase. Both are treated as red flags regardless of the quality of the final work.

3. Written Reflection (End of Project Report): (3 marks)

The project report should include a Reflection section (300–500 words) as the last section. This section must be written in students' own words and should address some of the following:

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method/model over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

Marking Criteria

Component	Suggested Weight
Final report/deliverable	15
Mid Progress report	2
Version/commit history	Pass/fail
Reflection section	3

Mid project Progress Report

Project Title: Book Inquiry System

Platform: GitHub & Visual Studio Code

1. Work Completed This Period:

- Examined the requirements of the Book Inquiry System and created a modular structure for the Java packages.
 - Setup: Installed and set up OpenJDK, VS Code, and Git Bash in the computer of group members.
 - Version Control Setup: Created remote repository in GitHub and configured remote tracking branches and collaborative work.
 - Core Development : Developed core java source code structure for following three files:
 - 1-Book.java: Created data variables (title, author, isbn), constructor, getter methods, and toString method.
 - 2-SearchSolutions.java: Developed (ArrayList sequential linear search) and (HashMap direct search by key).
 - 3-Main.java: Created the execution source code to create the test dataset and print runtime of nanoseconds ($O(n)$ vs $O(1)$).
- Documentation and Teamwork: Created and published README.md with setup details and individual responsibility of each member and Big-O complexity table. Successfully merged author documentation commit from all group members.

2. One key decision has been made during this period:

- Different files: Split the domain object (Book), algorithmic logic (SearchSolutions), and execution driver (Main) into different files to enable development with team members so every member can edit their files freely.

3. One problem encountered:

Git Username and installation: The Problem was git couldn't detect email address or username when attempting to commit in local git so syncing was impossible.

4. Plan for Next Phase:

1. Increase the number of entries in Main.java file to over 1,000 entries to check execution speed.
2. Improve search capability of SearchSolutions.java file by including input and case-insensitive searches.
3. Commit on all members laptops.

5. Each group member input/participation:

- **ABDULLAH ALOFAIDLI (Team Lead):**
 - Set up the Git/GitHub repository infrastructure and README.md documentation.
 - Implemented the execution driver and performance benchmarking loop (Main.java).
 - Dealt with git rebase conflicts (local/remote) and branch merging.
- **Mohammed alqahtani (Data Model Developer):**
 - Book.java class fields, constructors and getters organized.
 - Documented the data model architecture of Book.java and pushed it to GitHub.
- **Ahmed Alzahrani (Algorithm Developer):**

- Implemented SearchSolutions.java with $O(n)$ ArrayList and $O(1)$ HashMap search solutions.
- Wrote and pushed search method documentation to GitHub.

Project Artifacts:

- GitHub/GitLab repository (with commit history):
<https://github.com/abdullah-115se/Book-inquiry-system-IT-245-PROJECT->
- Live report document (with edit history):
https://docs.google.com/document/d/1gd7_tckRW_aJ4IJCr6PsDxQDqnXWuI-6NRCgXszgIEI/edit?tab=t.0